

IR PROTOCOL ANALYSER

Here we discuss a method to decode the IR signal easily using very few resources. Anyone who has access to a personal computer can try it

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Nowadays use of an IR remote control (IRC) as a substitute for a keypad is becoming popular among microcontroller-based project developers and hobbyists, as a large number of keys are available on the remote control compared to the conventional keypad. An IR receiver like TSOP17XX that uses only one pin of a microcontroller can be employed to decode the IR signal, whereas a keypad consumes eight pins for a 4×4 keypad (16-key). Also, on a remote control used for TV, you can find nearly 30 keys.

Finally, you get a neat and elegant look for your project.

Of course, there are difficulties too associated with using an IRC. The first is writing the code for decoding the signal and identifying the key pressed. The other is identifying the command sent by a key. If you don't have prior knowledge about the command corresponding to the key you intend to use, things become more complicated. This, however, becomes a problem only if you are using a large number of keys because you can find lots of resources in the Internet that provide information about various remote controls and the commands of common keys found. You can't decode the signal using a CRO because the signal isn't continuous and the only way out is to use a DSO.

Here we discuss a method to decode the IR signal very easily using few resources. Anyone who has access to a personal computer can use this project. Resources required are:

1. A PC with sound card and microphone input
2. Windows operating system
3. Sound recorder software (provided with Windows)
4. Nero WaveEditor software or a similar one

IR signals

Before going into the project, it is necessary to understand the IR signals from an IRC. IR radiation isn't visible, as it lies outside the visible spectrum. IRCs use IR signals to transmit information from the handheld unit to the remote-operated system. In fact, everything that radiates heat also radiates IR. Therefore we modulate the signal and then transmit it to make sure that the receiver gets the IR message that we send. The receiver is tuned to the carrier frequency and therefore rejects all other IR signals from various sources.

Every key on the remote control has an address and command which are sent as modulated digital pulses when the

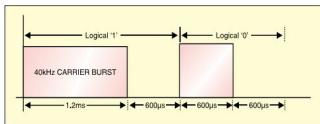


Fig. 1: Modulated signal

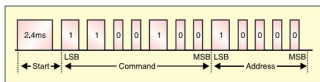


Fig. 2: Pulse train of the SIRC protocol

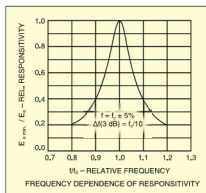


Fig. 3: Frequency response of the photo module

corresponding key is pressed. Address is unique for all keys on the same remote control, while command is different for each key.

Here we will discuss the SIRC protocol that is popular among hobbyists.

SIRC. SIRC

stands for 'Sony infrared remote control.' It features:

1. 12-bit long sequence (commonly found)
2. 5-bit address and 7-bit command length (5 bits + 7 bits = 12 bits)
3. PWM
4. Carrier frequency of 40 kHz
4. Bit time of 1.2 ms or 0.6 ms

Modulated signal. The signal sent is a burst of 40kHz carrier and a 1.2ms long burst represents a logic '1' and a 0.6ms long burst represents a logic '0'. All bursts are separated by a 0.6ms long interval.

Pulse train. Fig. 2 shows a pulse train of the SIRC protocol. The least significant bit (LSB) is transmitted first. The start burst is 2.4ms wide, followed by a standard space of